

EDUCATION

Stanford University <i>Ph.D. in Computer Science</i> <i>Advised by Chris Manning and Chris Potts</i>	<i>2022 - 2026</i>
Stanford University <i>M.S. in Symbolic Systems Program</i>	<i>2020 - 2022</i>
University of Pennsylvania <i>M.S. in Computer Science</i>	<i>2015 - 2017</i>
Case Western Reserve University <i>B.S. in Aerospace Engineering</i>	<i>2012 - 2015</i>

MANUSCRIPTS AND PUBLICATIONS¹

- preprint* XYBENCH: AGENTS ARE YES-MEN TO BAD ADVICE
Zhengxuan Wu*, Yuxuan Li*, Oyvind Tafjord, Been Kim
https://zen-wu.social/papers/xybench_preview.pdf.
- preprint* THE PIGGYBACK HYPOTHESIS OF GENERALIZATION: EXPLAINING AND MITIGATING EMERGENT MISALIGNMENT
Jiahao Zhao, **Zhengxuan Wu**, Aryaman Arora, Yanchao Sun, David Bau, Weiyan Shi
<https://arxiv.org/abs/2606.06667>.
- preprint* PREFT: PREFILL-ONLY FINETUNING FOR EFFICIENT INFERENCE
Anthony Lanpouthakoun, Aryaman Arora, **Zhengxuan Wu**, Dhruv Pai, Benjamin Keigwin, Dan Jurafsky, Christopher Potts
<https://arxiv.org/abs/2605.14217>.
- preprint* HYPERSTEER: ACTIVATION STEERING AT SCALE WITH HYPERNETWORKS
Jiuding Sun*, Sidharth Baskaran*, **Zhengxuan Wu**, Michael Sklar, Christopher Potts, Atticus Geiger
<https://arxiv.org/abs/2506.03292>.
- preprint* GIM: IMPROVED INTERPRETABILITY FOR LARGE LANGUAGE MODELS
Joakim Edin, Róbert Csordás, Tuukka Ruotsalo, **Zhengxuan Wu**, Maria Maistro, Casper L. Christensen, Jing Huang, Lars Maaløe
<https://arxiv.org/abs/2505.17630>.
- preprint* A REPLY TO MAKELOV ET AL.(2023)’S “INTERPRETABILITY ILLUSION” ARGUMENTS
Zhengxuan Wu, Atticus Geiger, Jing Huang, Aryaman Arora, Thomas Icard, Christopher Potts, Noah D. Goodman <https://arxiv.org/abs/2401.12631>.

¹*equal contribution

- CoLM '26 ADAG: AUTOMATICALLY DESCRIBING ATTRIBUTION GRAPHS
Aryaman Arora, **Zhengxuan Wu**, Jacob Steinhardt, Sarah Schwettmann
<https://transluce.org/neuron-circuits>.
- ICML '26 LANGUAGE MODEL CIRCUITS ARE SPARSE IN THE NEURON BASIS
spotlight Aryaman Arora*, **Zhengxuan Wu***, Jacob Steinhardt, Sarah Schwettmann
<https://transluce.org/neuron-circuits>.
- NeurIPS '25 IMPROVED REPRESENTATION STEERING FOR LANGUAGE MODELS
spotlight **Zhengxuan Wu***, Qinan Yu*, Aryaman Arora, Christopher D. Manning, Christopher Potts
<https://arxiv.org/abs/2505.20809>.
- NeurIPS '25 LLMs ENCODE HARMFULNESS AND REFUSAL SEPARATELY
Jiachen Zhao, Jing Huang, **Zhengxuan Wu**, David Bau, Weiyan Shi
<https://arxiv.org/abs/2507.11878>.
- Nature QUANTIFYING LARGE LANGUAGE MODEL USAGE IN SCIENTIFIC PAPERS
Human Weixin Liang, Yaohui Zhang, **Zhengxuan Wu**, Haley Lepp, Wenlong Ji, Xuandong
Behaviour Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher Potts,
'25 Christopher D. Manning, James Zou
<https://www.nature.com/articles/s41562-025-02273-8>.
- ICML '25 AXBENCH: STEERING LLMs? EVEN SIMPLE BASELINES OUTPERFORM SPARSE
spotlight AUTOENCODERS
Zhengxuan Wu*, Aryaman Arora*, Atticus Geiger, Zheng Wang, Jing Huang,
Dan Jurafsky, Christopher D. Manning, Christopher Potts
<https://arxiv.org/abs/2501.17148>.
- JMLR '25 CAUSAL ABSTRACTION: A THEORETICAL FOUNDATION FOR MECHANISTIC IN-
TERPRETABILITY
Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan,
Jing Huang, Aryaman Arora, **Zhengxuan Wu**, Noah Goodman, Christopher Potts,
Thomas Icard <https://arxiv.org/abs/2301.04709>.
- EMNLP '24 DANCING IN CHAINS: RECONCILING INSTRUCTION FOLLOWING AND FAITHFUL-
NESS IN LANGUAGE MODELS
Zhengxuan Wu, Yuhao Zhang*, Peng Qi*, Yumo Xu*, Rujun Han, Yian Zhang,
Jifan Chen, Bonan Min, Zhiheng Huang <https://arxiv.org/pdf/2407.21417>.
- NeurIPS '24 REFT: REPRESENTATION FINETUNING FOR LANGUAGE MODELS
spotlight **Zhengxuan Wu***, Aryaman Arora*, Zheng Wang, Atticus Geiger, Dan Jurafsky,
Christopher D. Manning, Christopher Potts <https://arxiv.org/abs/2404.03592>.
- CoLM '24 MAPPING THE INCREASING USE OF LLMs IN SCIENTIFIC PAPERS
Weixin Liang*, Yaohui Zhang*, **Zhengxuan Wu***, Haley Lepp, Wenlong Ji, Xuan-
dong Zhao, Haocheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher
Potts, Christopher D. Manning, James Y. Zou <https://arxiv.org/abs/2404.01268>.
- NAACL '24 PYVENE: A LIBRARY FOR UNDERSTANDING AND IMPROVING PYTORCH MODELS
VIA INTERVENTIONS
Zhengxuan Wu, Atticus Geiger, Aryaman Arora, Jing Huang, Zheng Wang, Noah
Goodman, Christopher D. Manning, Christopher Potts
<https://arxiv.org/abs/2403.07809>.

- CogSci '24 SYMBOLIC VARIABLES IN DISTRIBUTED NETWORKS THAT COUNT
Satchel Grant, **Zhengxuan Wu**, James Lloyd McClelland, Noah Goodman
<https://escholarship.org/uc/item/7gm9d3hp>.
- ICML '24 IN-CONTEXT SHARPNESS AS ALERTS: AN INNER REPRESENTATION PERSPECTIVE FOR HALLUCINATION MITIGATION
Shiqi Chen*, Miao Xiong*, Junteng Liu, **Zhengxuan Wu**, Teng Xiao, Siyang Gao, Junxian He <https://arxiv.org/abs/2403.01548>.
- ACL '24 RAVEL: EVALUATING INTERPRETABILITY METHODS ON DISENTANGLING LANGUAGE MODEL REPRESENTATIONS
Jing Huang, **Zhengxuan Wu**, Christopher Potts, Mor Geva, Atticus Geiger
<https://arxiv.org/abs/2402.17700>.
- Clearr '24 FINDING ALIGNMENTS BETWEEN INTERPRETABLE CAUSAL VARIABLES AND DISTRIBUTED NEURAL REPRESENTATIONS
Atticus Geiger*, **Zhengxuan Wu***, Christopher Potts, Thomas Icard, Noah D. Goodman, <https://arxiv.org/abs/2303.02536>.
- BlackboxNLP '23 RIGOROUSLY ASSESSING NATURAL LANGUAGE EXPLANATIONS OF NEURONS [**Best Paper Award**]
Jing Huang, Atticus Geiger, Karel D'Oosterlinck, **Zhengxuan Wu**, Christopher Potts, <https://arxiv.org/abs/2309.10312>.
- EMNLP '23 MQUAKE: ASSESSING KNOWLEDGE EDITING IN LANGUAGE MODELS VIA MULTI-HOP QUESTIONS
Zexuan Zhong*, **Zhengxuan Wu***, Christopher D. Manning, Christopher Potts, Danqi Chen, <https://arxiv.org/abs/2202.12312>.
- EMNLP '23 OOLONG: INVESTIGATING WHAT MAKES CROSSLINGUAL TRANSFER HARD WITH CONTROLLED STUDIES
Zhengxuan Wu*, Isabel Papadimitriou*, Alex Tamkin*,
<https://arxiv.org/abs/2202.12312>.
- TACL '23 RECOGS: HOW INCIDENTAL DETAILS OF A LOGICAL FORM OVERSHADOW AN EVALUATION OF SEMANTIC INTERPRETATION
Zhengxuan Wu, Christopher D. Manning, Christopher Potts,
<https://arxiv.org/abs/2303.13716>.
- NeurIPS '23 INTERPRETABILITY AT SCALE: IDENTIFYING CAUSAL MECHANISMS IN ALPACA
Zhengxuan Wu*, Atticus Geiger*, Thomas Icard, Christopher Potts, Noah D. Goodman, <https://arxiv.org/abs/2305.08809>.
- Findings ACL '23 INDUCING CHARACTER-LEVEL STRUCTURE IN SUBWORD-BASED LANGUAGE MODELS WITH TYPE-LEVEL INTERCHANGE INTERVENTION TRAINING
Jing Huang, **Zhengxuan Wu**, Kyle Mahowald, Christopher Potts,
<https://arxiv.org/abs/2209.14279>.
- ICML '23 CAUSAL PROXY MODELS FOR CONCEPT-BASED MODEL EXPLANATIONS
Zhengxuan Wu*, Karel D'Oosterlinck*, Atticus Geiger*, Amir Zur, Christopher Potts, M.s., Stanford University, <https://arxiv.org/abs/2209.14279>.
- NeurIPS '22 ZERO-C: A NEURO-SYMBOLIC MODEL FOR ZERO-SHOT CONCEPT RECOGNITION AND ACQUISITION AT INFERENCE TIME

- Tailin Wu, Megan Tjandrasuwita, **Zhengxuan Wu**, Xuelin Yang, Kevin Liu, Rok Sasic, Jure Leskovec,
<https://arxiv.org/abs/2206.15049>.
- NeurIPS '22 CEBAB: ESTIMATING THE CAUSAL EFFECTS OF REAL-WORLD CONCEPTS ON NLP MODEL BEHAVIOR
 Eldar David Abraham*, Karel D'Oosterlinck*, Amir Feder*, Yair Ori Gat*, Atticus Geiger*, Christopher Potts*, Roi Reichart*, **Zhengxuan Wu***,
<https://arxiv.org/abs/2205.14140>.
- ICML '22 INDUCING CAUSAL STRUCTURE FOR INTERPRETABLE NEURAL NETWORKS
 Atticus Geiger*, **Zhengxuan Wu***, Hanson Lu*, Josh Rozner, Elisa Kreiss, Thomas Icard, Noah D. Goodman, Christopher Potts, <https://arxiv.org/abs/2112.00826>.
- NAACL '22 CAUSAL DISTILLATION FOR LANGUAGE MODELS
Zhengxuan Wu*, Atticus Geiger*, Josh Rozner, Elisa Kreiss, Hanson Lu, Thomas Icard, Christopher Potts, Noah D. Goodman, <https://arxiv.org/abs/2112.02505>.
- RepL4NLP '22 IDENTIFYING THE LIMITS OF CROSS-DOMAIN KNOWLEDGE TRANSFER FOR PRE-TRAINED MODELS [**Best Paper Award**]
Zhengxuan Wu, Nelson F. Liu, Christopher Potts,
<https://arxiv.org/abs/2104.08410>.
- NeurIPS '21 REAScan: COMPOSITIONAL REASONING IN LANGUAGE GROUNDING
Zhengxuan Wu*, Elisa Kreiss*, Desmond C. Ong, Christopher Potts,
<https://arxiv.org/abs/2109.08994>.
- ACL '21 DYNASent: A DYNAMIC BENCHMARK FOR SENTIMENT ANALYSIS
 Christopher Potts*, **Zhengxuan Wu***, Atticus Geiger, Douwe Kiela,
<https://arxiv.org/abs/2012.15349>.
- NAACL '21 DYNABENCH: RETHINKING BENCHMARKING IN NLP'
 Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, **Zhengxuan Wu**, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Waseem, Pontus Stenetorp, Robin Jia, Mohit Bansal, Christopher Potts and Adina Williams, <https://arxiv.org/abs/2104.14337>.
- AAAI '21 CONTEXT-GUIDED BERT FOR TARGETED ASPECT-BASED SENTIMENT ANALYSIS
Zhengxuan Wu, Desmond C. Ong, <https://arxiv.org/abs/2010.07523>.
- CHI '21 NOT NOW, ASK LATER: USERS WEAKEN THEIR BEHAVIOR CHANGE REGIMEN OVER TIME, BUT BELIEVE THEY WILL IMMINENTLY RE-STRENGTHEN IT
 Geza Kovacs, **Zhengxuan Wu** and Michael S. Bernstein,
<https://arxiv.org/abs/2101.11743>.
- SCiL '21 PRAGMATICALLY INFORMATIVE COLOR GENERATION BY GROUNDING CONTEXTUAL MODIFIERS
Zhengxuan Wu, Desmond C. Ong, <https://arxiv.org/abs/2010.04372>.
- BlackboxNLP '20 STRUCTURED SELF-ATTENTION WEIGHTS ENCODE SEMANTICS IN SENTIMENT ANALYSIS
Zhengxuan Wu, Thanh-Son Nguyen and Desmond C. Ong,
<https://arxiv.org/abs/2010.04922>.

- ACII '19 ATTENDING TO EMOTIONAL NARRATIVES
Zhengxuan Wu, Xiyu Zhang, Zhi-Xuan Tan, Jamil Zaki, Desmond C. Ong,
<https://arxiv.org/abs/1907.04197>.
- TAC '19 MODELING EMOTION IN COMPLEX STORIES: THE STANFORD EMOTIONAL NARRATIVES DATASET
 Desmond C. Ong, **Zhengxuan Wu**, Zhi-Xuan Tan, Marianne Reddan, Isabella Kahhale, Alison Mattek and Jamil Zaki, <https://arxiv.org/abs/1912.05008>.
- CHI '19 CONSERVATION OF PROCRASTINATION: DO PRODUCTIVITY INTERVENTIONS SAVE TIME OR JUST REDISTRIBUTE IT?
 Geza Kovacs, Drew Mylander Gregory, Zilin Ma, **Zhengxuan Wu**, Golrokh Emami, Jacob Ray and Michael S. Bernstein, <https://dl.acm.org/doi/10.1145/3290605.3300560>.
- CSCW '18 ROTATING ONLINE BEHAVIOR CHANGE INTERVENTIONS INCREASES EFFECTIVENESS BUT ALSO INCREASES ATTRITION
 Geza Kovacs, **Zhengxuan Wu** and Michael S. Bernstein,
<https://dl.acm.org/doi/10.1145/3274364>.

OTHER PROFESSIONAL EXPERIENCE

- | | |
|---|--------------------|
| Google DeepMind - Senior Research Scientist | <i>2026 -</i> |
| · Developing methods for agent safety and interpretability. | |
| Transluce - Research Fellow | <i>2025 - 2026</i> |
| · Developed circuit tracing methods for language models. | |
| Meta, Inc. - Research Scientist Intern | <i>2024 - 2024</i> |
| · Developed interpretability tools for LLaMA models. | |
| Amazon, Inc. - Research Scientist Intern | <i>2023 - 2023</i> |
| · Investigated faithfulness and instruction following of language models. | |
| VMware, Inc. - Senior Member of Technical Staff | <i>2017 - 2022</i> |
| · Developed scalable data-center management platform. | |
| Swift Capital (Paypal, Inc.) - Machine Learning Intern | <i>2016 - 2016</i> |
| · Developed machine learning systems to predict the credit scores of loan applicants. | |

ACADEMIC EXPERIENCE

- Reviewer for CHI19, *CL22-24, ICML22-24, NeurIPS22-23, COLM24
- Invited Abstract Presentation in IC2S2 2019, University of Amsterdam, Netherlands

TECHNICAL STRENGTHS

- **Program Languages:** Python, C++/C, C#, Java, R, Matlab, Haskell, Bash.
- **Machine Learning:** Discriminative and Generative Models; Reinforcement Learning; Multi-task Learning; Graph Neural Networks.
- **AI + Big Data:** PyTorch, scikit-learn, Keras, TensorFlow, NumPy, Pandas, H2O, MapReduce (Hadoop).
- **Data Mining:** PyData, SciPy, SNAP, SQL, NoSQL (Mongo), NetworkX, Jupyter.
- **Data Science:** Mixed Linear Model, Hierarchical Logistic Regression, A/B Testings, Crowdsourcing (MTurk).
- **Server + Database:** Node.js, Flask, MongoDB, PostgreSQL, Kubernetes, Docker, Google Cloud, AWS EC2, Heroku, Azure, Jenkins CICD.

- **Web + Mobile:** HTML/CSS/JS, Polymer, React, Webpack, Apache, Android (Java), Xcode.